SIoV Mobility Management Using SDVN-Enabled Traffic Light Cooperative Framework

NEETESH KUMAR, Indian Institute of Technology Roorkee, Roorkee, India
NAVJOT SINGH, Indian Institute of Information Technology Allahabad, Prayagraj, India
ANUJ SACHAN, Indian Institute of Technology Roorkee, Roorkee, India
RASHMI CHAUDHRY, Netaji Subhas University of Technology, Delhi, India

Social Internet of Vehicles (SIoV) is an emerging connected vehicular networking framework among specialized social vehicles to share and disseminate important information like traffic updates, weather conditions, parking slots, so on. This study aims to form an SIoV network among emergency vehicles for their frequent communication to improve the throughput, average waiting time, queue length, and speed during vehicular movement while crossing the intersection in the city. To address this, we propose a novel Smart Traffic Light Controller (STLC)-assisted Software-Defined Vehicular Networking (SDNV)-enabled SIoV framework for emergency vehicles. Emergency vehicles form an SIoV network by utilizing SDVN architecture in vehicle-to-vehicle and vehicle-to-infrastructure communication. The SDVN module is used to offer two essential services: (1) SIoV-based road-lane prioritization and (2) congestion prevention signal generation for the STLC. An SDVN-MP algorithm is proposed to generate an effective traffic light control signal with an SDVN controller feedback signal. Furthermore, to improve the SIoV movement in the city, two levels of prioritization are done: (1) SIoV and (2) the road lane with SIoV. The first level of prioritization is to assign higher weightage to the social vehicular entities, and the second level is to prioritize the respective road lane based on SIoV quantity. The proposed framework is validated through a realistic simulation study on the Indian city OpenStreetMap utilizing the Simulation of Urban MObility simulator. The experimental findings demonstrate that the SDVN-MP model enhances (state-of-the-art) comparative performance by 22.5–55.2%, 1.2–82.7%, 1.6–38.4%, and 1.8–12.4% for average waiting time, average speed, average queue length, and average throughput metrics, respectively.

CCS Concepts: • **Networks** → **Programmable networks**; • **Computer systems organization** → **Sensors and actuators**;

Additional Key Words and Phrases: Social Internet of vehicles, software-defined vehicular networking, traffic light controller, scheduling.

This work was supported by Science and Engineering Research Board (SERB), Government of India under Grant EEQ/2019/000182, and the Council of Scientific and Industrial Research (CSIR), New Delhi, Government of India under Grant 22-0817-19-EMR-II.

Authors' Contact Information: Neetesh Kumar, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India; e-mail: neetesh@cs.iitr.ac.in; Navjot Singh, Indian Institute of Information Technology Allahabad, Jhalwa, Prayagraj, Uttar Pradesh, India; e-mail: navjot@iiita.ac.in; Anuj Sachan (Corresponding author), Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India; e-mail: anuj_s@cs.iitr.ac.in; Rashmi Chaudhry, Netaji Subhas University of Technology, New Delhi, Delhi, India; e-mail: rashmi@nsut.ac.in.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 2378-9638/2024/7-ART27
https://doi.org/10.1145/3653721

ACM Reference format:
Neetesh Kumar, Navjot Singh, Anuj Sachan, and Rashmi Chaudhry. 2024. SIoV Mobility Management Using SDVN-Enabled Traffic Light Cooperative Framework. *ACM Trans. Cyber-Phys. Syst.* 8, 3, Article 27 (July 2024), 24 pages.
https://doi.org/10.1145/3653721

1 Introduction

Vehicular technology has been enriched with sensing, **Internet of Things (IoT)**, and computing & communication technologies-enabled smart devices, i.e., **Road Side Unit (RSU)** and **On Board Unit (OBU)**, Intelligent Vehicles, and connected **Traffic Light Controllers (TLCs)** [18]. The traditional **Vehicular *Ad-hoc* Networks (VANETs)** model, along with RSUs and OBUs, are being transformed into the **Social Internet of Vehicles (SIoV)** [5]. SIoV is the modern trend toward IoV, where entities socialize with each other by sharing information about common interests. SIoV has given rise to several intelligent services such as on-road traffic information, gas stations, nearby restaurants, hospitals, and lots of other utility services [3, 14, 16, 37]. Technically, an SIoV network is a dynamic vehicular network to address the transport safety issue among a particular class of vehicles utilizing **Vehicle-to-Internet (V2I)**, **Vehicle-to-Vehicle (V2V)**, **Vehicle-to-Infrastructure (V2X)**, battery-powered intelligent devices, and Cloud/Fog servers [32]. In brief, SIoV is an embryonic vehicular cyber-physical system that can extend VANETs, IoT, and Fog or Edge servers collectively [6]. In this work, the problem of emergency vehicles (ambulance, fire brigade, etc.) movement with minimal congestion is tackled by forming an emergency SIoV network and utilizing an efficient **Software-Defined Vehicular Networking (SDVN)** architecture. Emergency SIoV is a SIoV network, particularly formed among emergency vehicles. Further, it is also observed that to offer less-congested and smoother transport services at the intersection for SIoV, a **Smart Traffic Light Controller (STLC)** plays a vital role in making the traffic flow scheduling decision. Thus, this work offers a solution for emergency SIoV by proposing an SDVN-enabled STLC cooperative SIoV architecture.

Traditional centralized architectures have limitations in scalability and are vulnerable to single points of failure, which requires the maintenance of expensive backups in non-virtualized infrastructure [8]. To address these issues, decentralized architectures have been explored [9]. However, decentralized architectures not only offer scalability but also have high operational, communication, maintenance, and management costs due to a lack of programmable control and synchronization overheads [9]. Additionally, the collaborative decision-making process in decentralized architectures experiences significant delays for large-scale deployments due to synchronization overhead [7]. Therefore, neither decentralized nor centralized control architectures are sufficient to provide a manageable, scalable, and cost-effective solution. To overcome the limitations of traditional architectures, a new paradigm called SDVN architectures is utilized as the network archetype in vehicular communication networks. SDVN architectures offer global coverage, low maintenance, programmable control interfaces, virtual infrastructure deployment, and auto-configurable features [25][26]. Furthermore, SDVN-based architectures can also be deployed as hybrid architectures to leverage the complementary advantages of both types of architectures [24].

In the same line, Sadio et al. [27] introduced an architecture composed of four layers: Cloud, SDVN controller, Fog, and vehicles to offer programmable control, scalability, flexibility, and global knowledge. The SDVN-enabled control system has also been proposed for traffic management considering emergencies in smart cities [4][17]. For SIoV, Xia et al. [34] introduced an incentive scheme for Fog computing-based SIoV network utilizing 2-dimensional behavior marker-based data. The authors evaluated the behaviors of vehicular nodes and promoted the cooperation of

data-forwarding among selfish nodes. Recently, Li et al. [20] have proposed an emergency information diffusion model for SIoV to disseminate information regarding emergency events to vehicles. From this study, it is observed that existing technologies such as VANETs, V2V, V2I, and V2X leverage SIoV. However, the following state-of-the-art research gaps need to be addressed for emergency SIoV network: (1) The existing SIoV network has been explored using Fog/Cloud; however, SIoV needs to be explored using SDVN archetype for most prioritized emergency vehicles in heterogeneous vehicular networks. (2) Existing methods lack in generating a "congestion prevention signal" by detecting the emergency SIoV at the neighboring intersections' outgoing links in advance, which causes the delay in making an efficient traffic flow control decision. To the best of the authors' knowledge, the SIoV networks still need to be explored with traffic light cooperative SDVN architecture to solve the congestion problem at the intersections. Therefore, this work proposes an SDVN-enabled and TLC-assisted SIoV framework to bridge the gap from the research mentioned for emergency services in the city. The significant contributions of the article can be summarized in the following folds:

— Traffic light cooperative SIoV- aware software-defined networking-enabled architecture is proposed by decoupling vehicular networking architecture into data and control planes. The proposed framework is novel due to the two-level prioritization: SIoV and the respective road lane with SIoV. Further, the SDVN module offers two essential services: road lane with SIoV prioritization and congestion prevention signal generation for the STLC.
— A vehicle weight assignment scheme is proposed to assign separate weights for the SIoV and heterogeneous vehicles to estimate an effective queue length instead of normal vehicle counts.
— An algorithm is proposed to tackle the congestion in the city. The **Application Control Server (ACS)** periodically monitors the traffic congestion in the city and generates the congestion prevention signal when congestion is detected at the respective intersection.
— An algorithm for SDVN controller is also proposed to increase the optimal priority of the lane when SIoV is detected. Further, an SDVN-**max-pressure (MP)** algorithm is proposed to generate an effective traffic light control signal with an SDVN controller feedback signal.
— Finally, an extensive experimental study is carried out using a well-known open-source simulator **Simulation of Urban MObility (SUMO)**. Rigorous experimental analysis validates the significance of each of the modules. Furthermore, the comparative experimental study proves the effectiveness of the proposed SDVN-feedback-aware MP algorithm over other state-of-the-art approaches.

The rest of the article is organized as follows. Section 2 critically explores the related literature on SIoV. Section 3 presents the details of the problem statement. Section 4 presents an SDVN-SIoV framework. Section 5 consists of the simulation study and a case study discussion with an in-depth analysis of experimental results. Finally, Section 6 concludes the article and provides suggestions for future work.

2 Literature Survey

The **Intelligent Transportation System (ITS)** has emerged as the principal focus of smart city research [31]. SIoV has been suggested as an efficient paradigm to offer safety and manage traffic, intelligent traffic surveillance, disasters, and other emergency services [5]. In SIoV, one entity needs to be connected with numerous entities, necessitating that the use of an effective architecture is necessarily required [14]. With this vehicular network's dynamic requirements, **Software-Defined SIoV (SD-SIoV)** has been progressing as a network archetype in the ITS to enable the vehicular network with dynamic, reliable, robust, and auto-configurable features. SD-SIoV has also been proposed for wireless networks to improve tractability and adaptability for network setups for

exponentially growing traffic [33]. Few energy-efficient routing in stochastic situations, SD-SIoV architecture, has also been proposed in light of the capability of SD-SIoV to manage vehicular networks with multiple benefits [16].

To adapt the advantages of SD control networks in ITS, a **Software-Defined Networking (SDN)**-based control system was developed for smart traffic control and management in urban emergencies [32]. However, the authors neglected the congestion caused by regular traffic on the road lanes. In recent, Balta et al. [2] presented a model for traffic signal optimization in large cities based on an SDN-based VANET and a three-stage fuzzy-decision tree, but the authors overlooked traffic heterogeneity and load balancing in congested regions of the city. Elise et al. [30] proposed a novel reward for coordinated **Deep Reinforcement Learning (DRL)** that considered the penalty function of waiting time, delay, crashes, jams, and deceleration. However, the researchers ignored the priority among the vehicles while training the agent. Also, there is no congestion monitoring in real-time, and there are no explicit congestion control signals. Jutury et al. [12] proposed a framework based on fuzzy logic to generate traffic light control signals considering heterogeneous traffic light scenarios. Calculation of the duration for the traffic light phase helps in improving various performance matrices such as **Average Waiting Time (AWT)**, **Average Traffic Flow (ATF)**, **Average Queue Length (AQL)**, and **Average Speed (AS)**. Liu et al. [22] demonstrated the Q-learning algorithm with redefined action spaces and reward functions enhances system performance, particularly in complex traffic scenarios. Additionally, the system determines all feasible phase durations through optimal action space. Sachan et al. [25] proposed a quantized MP-based TLC using SDN to improve the traffic flow at the junctions and avoid starvation situations at the intersection. However, no explicit focus on SIoV is given. Sun et al. [28] deployed a model for predicting the traffic event using representation learning. Chu et al. [10] proposed an end-to-end off-policy DRL approach for traffic light signal control to mitigate traffic congestion. Kumar et al. [17] proposed a novel traffic-light scheduling framework utilizing DRL and SD control to optimize traffic flow and prevent congestion in urban areas but ignored the SIoV. Sachan et al. [26] proposed an SDN-enabled quantized linear quadratic regulator to reduce congestion at intersections. This work developed a multi-input and multi-output system capable of managing complex traffic scenarios using a SD control to enable real-time traffic monitoring and congestion mitigation. However, the study did not explicitly focus on the SIoV. Yazdani et al. [35] proposed a DRL adaptive traffic signal model considering vehicle and pedestrian flows. The study aimed to minimize "total user delays" rather than just "total vehicle delays." However, this study did not consider explicit mechanisms to avoid congestion or SIoV prioritization. Table 1 depicts the summary of the literature review and highlights the comparative features of the proposed framework.

From the above study, it is inferred that SDN architecture is beneficial in several aspects of the transport cyber-physical systems. However, present approaches fail to produce an explicit mechanism to avoid congestion by predicting emergency SIoV vehicles at adjacent intersections' outgoing links, resulting in delays in effective traffic flow management decisions. Notably, to the best of our knowledge, no SIoV networks have integrated a traffic light cooperative SDVN architecture to alleviate congestion concerns at intersections. It is also observed that TLCs play a significant role in smooth traffic flow. Therefore, considering the facts mentioned earlier and the state-of-the-art [36], this work proposes an SDN-enabled SIoV framework for emergency vehicle movement with minimal congestion in the city, with the appropriate cooperation of a TLC.

3 Problem Statement

In this work, we identified a problem of emergency vehicle movement on congested road lanes. We tackled it by proposing an SIoV framework with the SDVN controller feedback-enabled smart traffic light scheduling and congestion prevention algorithms where prioritization plays an important

Table 1. Literature Summary

Reference	Objective & Method	Advantage	Limitation	Heterogeneity	Simulation Tool
Kumar et al. [16]	To minimize the delay & two-way particle swarm optimization	Collaborative decision making, programmable control	Actuated based, SIoV prioritization	No	MATLAB
Balta et al. [2]	To minimize delay and CO_2 emission & ant colony optimization	Beneficial in dense and variable traffic conditions	Collaborative decision making	No	SUMO
Elise et al. [30]	To minimize the average travel time and standard error & DRL	Deal with complex traffic scenarios efficiently	Explicit mechanism to avoid congestion, collaborative decision making	No	SUMO
Jutury et al. [12]	To improve AWT, ATF, AQL, and AS & neuro-fuzzy	Multi mode execution, actuated based	Collaborative decision making	Yes	SUMO
Liu et al. [22]	To minimize AWT and AQL & deep Q-learning	Redefined action spaces and reward functions in complex traffic scenario	Explicit mechanism to avoid congestion	No	SUMO
Sachan et al. [25]	To improve AWT, AQL, AS, and throughput (TP) & max-pressure	Explicit mechanism to avoid congestion, programmable control	SIoV prioritization	Yes	SUMO
Chu et al. [10]	To minimize waiting time & DRL	Developed off-policy DRL agent integrated with background removal residual networks	Explicit mechanism to avoid congestion	No	SUMO
Kumar et al. [17]	To improve AWT, TP, AQL, and AS & DRL utilizing SDN	Explicit mechanism to avoid congestion, programmable control	Collaborative decision making, SIoV prioritization	Yes	SUMO
Sachan et al. [26]	To improve AWT, AQL, TP, and AS & linear quadratic regulator	Developed a complex traffic scenario-capable multi-input and multi-output system	Actuated based, SIoV prioritization	Yes	SUMO
Yazdani et al. [35]	To minimize total vehicle delays & DRL	Developed a DRL-based solution considering vehicle and pedestrian flows	Explicit mechanism to avoid congestion, SIoV prioritization	No	VISSIM
Proposed work	To improve *AWT, ATP, AQL, ALS & heuristic*	*Actuated based, collaborative decision making, explicit mechanism to avoid congestion, programmable control, SIoV prioritization*	-	*Yes*	*SUMO*

role for emergency vehicles. The explored problem is challenging due to real-time communication, vehicles' timely arrival requirements, heterogeneity among vehicles, uncertain vehicular dynamics, dense traffic light junctions, and frequent unexpected congestion on the road lanes. On the other hand, the TLC plays a significant role in the city's vehicle movement. An ineffective TLC increases vehicle waiting time and queue length at the intersections. SDVN controller-based network makes real-time communication among the vehicles feasible and effective. Therefore, a TLC cooperation-based SDVN-enabled SIoV framework is required to tackle this problem. Usually, a combination of the TLCs generates three signals: green, red, and yellow. When vehicles are moving in multiple directions to control the flow, at such traffic light intersections, the traffic lights change so that the traffic flows in various directions without interfering. The period spent at a single signal, such as green, red, or yellow, is called phase duration. At each junction, all phases shift cyclically, which is a cycle that repeats itself. No vehicle can proceed through an intersection that has a red traffic signal. On the other hand, if the traffic signal is green, the vehicle can proceed freely. The aims of this work are listed as follows: (1) to identify the emergency vehicles and create an SDVN-enabled SIoV architectural network, (2) to minimize the waiting time queue length and maximize the throughput and speed of the traffic flow at the intersections where emergency vehicles travel, and (3) to detect the relatively highly congested intersections in the city where the SIoV travels and to prevent the congestion by offering the congestion prevention signal for the STLC to minimize the waiting time.

4 The Proposed SDVN-SIoV Framework

4.1 SIoV System Model

The proposed model consists of SDVN-enabled SIoV architecture which decouples the vehicular network into data plane and control plane [19] as shown in Figure 1. The proposed model describes the SIoV with already established acronyms in the VANETs model, i.e., OBUs, and RSUs [1]. The OBUs are installed in emergency vehicles lie in data plane referred to as **Normal Nodes (NNs)**, and RSUs represent the roadside infrastructures and interconnected with SDVN controllers and OBUs using the Internet and Local area network (V2X), respectively. The SDVN controllers are referred to as **Control Nodes (CNs)**. It is defined as a control decision making node which processes data received from NNs and disseminates the control signals to the STLC. NN is defined as the data forwarding node [8] such as RSU/OBU. The externally powered ACS can be deployed over a Cloud/Fog server. It enables network control to be directly programmable and the customized abstraction of underlying infrastructure based on traffic applications and network services [32]. The CN acts as data aggregating (from NNs), data disseminating (to NNs), and data forwarding entity (to the **Control Server (CS)**) following an appropriate route. The NN is responsible for sending the data packet as a service request to the CN and receiving the service from the CN. The network is denoted with the help of a directed graph **G = (V,E)**, where **V** represents the set of intersections of road lanes (nodes) and **E** represents the set of road lanes (links). Each road lane is represented as an ordered pair **(u,v)**ϵ**E**. The set of input nodes and output nodes over **(u,v)** are represented by $In_{u,v}$ and $Ot_{u,v}$, respectively. In this scenario, the nodes of the network are classified into three classes- (1) source nodes, where vehicles enter the network, (2) destination nodes, where traffic flows out of the network, and (3) intermediate nodes, represented as the signalized intersection between source and destination nodes. This article proposes a traffic control system where optimized green, yellow, and red lights are generated at intermediate nodes while reducing the congestion. The model is discrete event-triggered where at link **(u,v)** the number of vehicles can be updated as

$$count_{u,v}(t+1) = count_{u,v}(t) + (\beta^{in}_{u,v}(t) - \beta^{out}_{u,v}(t)).T_{cl} \tag{1}$$

where t is the variable representing discrete-time index for the traffic network model and its controller. The β_{in} and βout are average incoming and outgoing flow, respectively. Here, T_{cl} represents

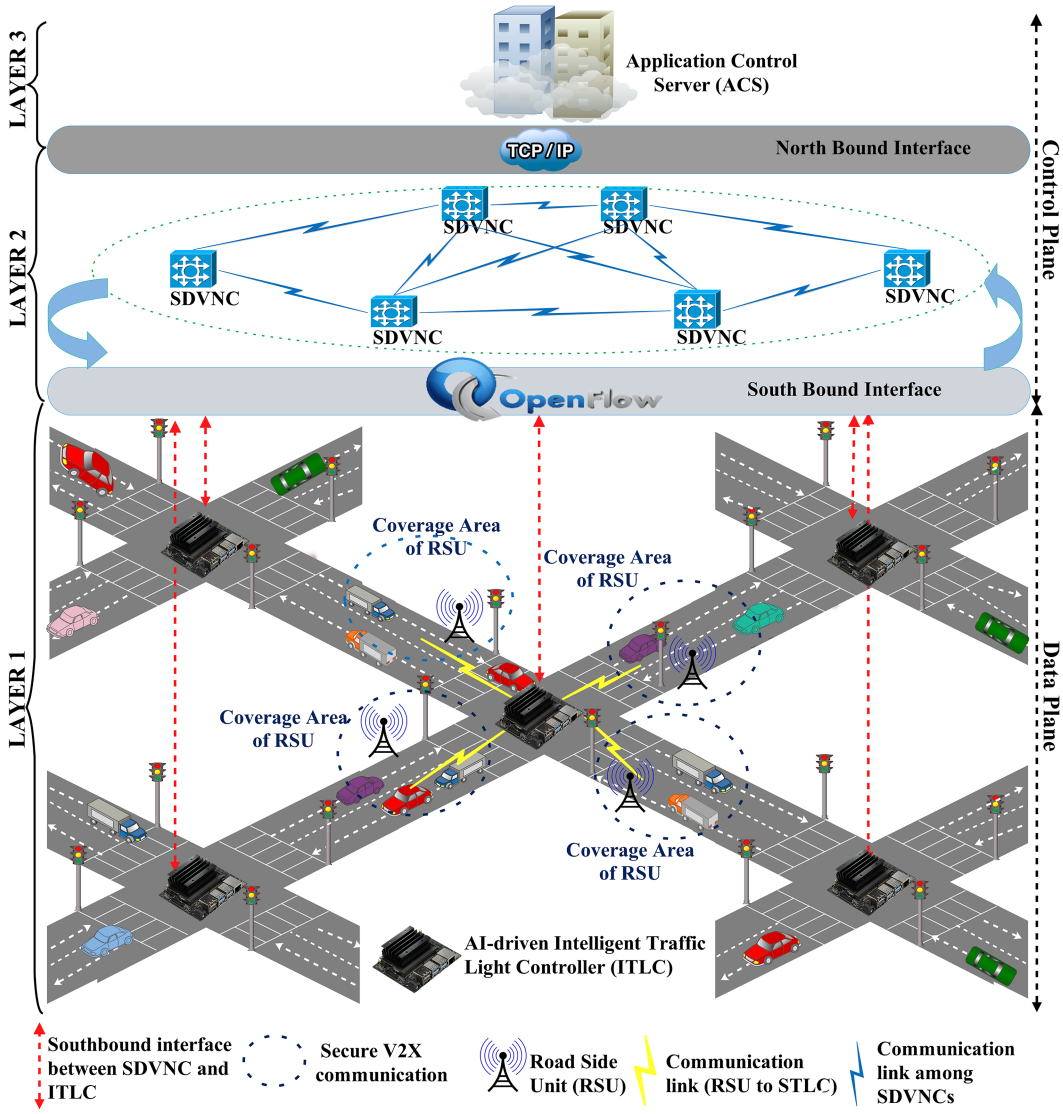


Fig. 1. Cyber-physical world interactions in SIoV with SDVN environment.

the traffic light control signal's cycle length at the intersection, and initially, that is constant for each intersection. Later, it is optimized online via the SDVN controller. To represent the actual scenario, incoming traffic flow and outgoing traffic flow, in SDN-based environment, has to be modeled to avoid congestion. In Equation (2), incoming and outgoing average flow for lane u,v is defined, where $\beta^{out}_{u,v,o}$ represents total flow at intermediate traffic junction o. Using this, the average flow which leaves and enters the link **(u,v)** is modeled as

$$\beta^{out}_{u,v}(t) = \sum_{o \epsilon Ot_{u,v}} \beta^{out}_{u,v,o}(t), \quad \beta^{in}_{u,v}(t) = \sum_{i \epsilon In_{u,v}} \beta^{out}_{i,u,v}(t) \tag{2}$$

where

$$\beta^{out}_{u,v,o}(t) = min\Big(a_{u,v,o}(t), b_{u,v,o}(t), c_{u,v,o}(t)\Big) \tag{3}$$

$$a_{u,v,o}(t) = \gamma_{u,v,o}(t) * \mu_{u,v} * g_{u,v,o}(t)/T_{cl} \tag{4}$$

$$b_{u,v,o}(t) = q_{u,v,o}(t)/T_{cl} + \beta^{arrive}_{u,v,o}(t) \tag{5}$$

$$c_{u,v,o}(t) = \frac{\gamma_{u,v,o}(t) \times (C_{v,o} - n_{v,o}(t))}{T_{cl} \times \sum\limits_{i \epsilon In_{v,o}} \gamma_{i,v,o}(t)} \tag{6}$$

where $a_{u,v,o}(t)$, $b_{u,v,o}(t)$, and $c_{u,v,o}(t)$ denote average flow rate for traffic conditions of saturated, unsaturated, and over-saturated scenarios, respectively. These three scenarios represent more real-time traffic needs which might exceed the available capacity of the system leading to congestion. For saturated traffic, a maximum flow which can be handled by a junction is dependent on four parameters: demand, cycle time, saturation flow, and effective green time duration [29]. Additionally, $\gamma_{u,v,o}(t)$ represents the part of traffic which turns from **(u,v)** to **(v,o)** via external input. Moreover, $g_{u,v,o}(t)$ denotes the green signal duration of the traffic stream going from **(u,v)** toward o at timestamp t, $\mu_{u,v}$ represents the saturated flow rate which leaves **(u,v)**, and the capacity of the link **(v,o)** is expressed as $C_{v,o}$.

As shown in Figure 1, the SDVN-SIoV framework is divided into three layers: (1) data plane layer, (2) control plane layer, and (3) ACS layer. Layer 1 and layer 2 are interfaced with the Application North Bound Interface, and Layer 2 and layer 3 are connected by South Bound Interface. Further, the SDVN-SIoV is partitioned into the control plane and the data plane. The control plane comprises r number of SDVN nodes and an ACS, and the data plane includes s number of RSUs/OBUs nodes referred to as NNs, where $n = r + s + 1$ is the count of the number of nodes in the SDVN-SIoV framework. In SDVN-SIoV, SDVN_{set} is the set of SDVN controllers, i.e., SDVN_1, SDVN_2, SDVN_3,...., $\text{SDVN}_r \epsilon \text{SDVN}_{set}$, and NN_{set} is the set of NNs, i.e., RSU_1, RSU_2, RSU_3,...., $\text{RSU}_{s_1} \epsilon \text{NN}_{set}$, OBU_1, OBU_2, OBU_3,...., $\text{OBU}_{s_2} \epsilon \text{NN}_{set}$ and $s = s_1 + s_2$. The first layer includes SIoV with deployed OBUs, or connected with RSUs at the transport infrastructure. SIoV/vehicles with deployed OBUs are connected with the RSUs/other vehicles based on their network coverage at the first layer. The ith SIoV is preferred to be connected with nearest jth RSU (using V2X network) based upon the distance d_{SIoV_i,RSU_j} computed as

$$d_{SIoV_i,\text{RSU}_j} = \sqrt{(x_{SIoV_i} - x_{\text{RSU}_j})^2 + (y_{SIoV_i} - y_{\text{RSU}_j})^2}, \tag{7}$$

where j is chosen as

$$j = argmin_{k=1..s} \quad d_{SIoV_i,\text{RSU}_k} \tag{8}$$

such that $d_{SIoV_i,\text{RSU}_k} < tr_{SIoV_i}$, where (x_{SIoV_i}, y_{SIoV_i}) and $(x_{\text{RSU}_j}, y_{\text{RSU}_j})$ are the coordinates of ith SIoV and jth RSU, respectively, and tr_{SIoV_i} is the transmission range of the ith SIoV. If the nearest RSU is not within the transmission range, then ith SIoV connects to the neighboring kth vehicle V_k (using V2V network), which is within the transmission range of ith SIoV and nearest to the jth RSU.

$$k = argmin_{V_l \in Neighbor_{SIoV_i}} \quad d_{V_l,\text{RSU}_j} \tag{9}$$

such that $d_{SIoV_i,V_k} < tr_{SIoV_i}$.

Following the above estimation about the transmission range and distance, the second layer includes SDVN controllers connected with the data plane layer's entities using SDVN switches with a 4G/5G network, and SDVN runs the application services from ACS. The third layer deploys

applications on the ACS and is remotely connected with the SDVN controllers, as shown in Figure 1. ACS is at a distant location from the SDVN controllers and other NNs. All SIoVs send their location information and ID and type to the nearby RSUs utilizing V2V or V2X networks.

STLC (Raspberry Pi/Nvidia Jetson) executes the proposed deployed traffic light scheduling algorithm based on the received data from RSUs/cameras/sensors and SDVN controllers for generating effective traffic light control signals. SDVN controller forms the clusters of RSUs/STLCs based on the distance.

The jth RSU is connected with nearest pth SDVN based upon the distance $d_{\text{SDVN}_p,\text{RSU}_j}$ computed as

$$d_{\text{SDVN}_p,\text{RSU}_j} = \sqrt{(x_{\text{SDVN}_p} - x_{\text{RSU}_j})^2 + (y_{\text{SDVN}_p} - y_{\text{RSU}_j})^2} \tag{10}$$

where p is chosen as

$$p = argmin_{q=1..r} \quad d_{\text{SDVN}_q,\text{RSU}_j} \tag{11}$$

such that $d_{\text{SDVN}_p,\text{RSU}_j} < tr_{\text{RSU}_j}$, where $(x_{\text{SDVN}_p}, y_{\text{SDVN}_p})$ and $(x_{\text{RSU}_j}, y_{\text{RSU}_j})$ are the coordinates of pth SDVN and jth RSU, respectively, and tr_{RSU_j} is the transmission range of the jth RSU. Each RSU/STLC connects with the nearest SDVN controller. Following this, SDVN controllers receive the data from the RSUs and process it to determine the SIoV and congestion on the road lanes. Next, processed data are sent to ACS to make control function decisions. ACS receives the data from each local SDVN controller and decides the lane priority and congestion prevention signal. The same is forwarded to the respective SDVN controllers. The SDVN controller forwards the respective signal to the STLC for further necessary actions. The STLC executes the traffic light scheduling algorithm by taking the received control signal from the SDVN controller as feedback.

4.2 Logical Flow

Figure 2 briefs about the proposed traffic light control-assisted SDVN-SIoV framework. From Figure 2, it can be visualized that the proposed framework is composed of two components: (1) STLC and (2) SDVN. STLC component begins by considering real-time traffic extracted vehicular dynamics as the inputs through RSUs/sensors/cameras deployed at each traffic light junction. **Real-time Traffic Parameter Extractor (RTPE)** module extracts required vehicular parameters such as **vehicle's ID (VID)**, AWT, AQL, AS, and **Average Throughput (ATP)**. Next, these parameters are sent to the lane heterogeneous weight calculator module to calculate the effective weight of each road lane by accounting for the heterogeneity among the vehicles. At the same time, all parameters are sent to the message send/receive module, which lies at the SDVN component to exchange the data packets between STLC and SDVN components. Based on traffic parameters, the traffic current state is also estimated by the RTPE module and forwarded to the STLC module, which is responsible for the successful execution of the traffic light control algorithm to generate a red/yellow/green signal.

At the same time, in the SDVN component, the message send/receive module analyzes the received traffic data packets and classifies them into two cases. For case 1, VIDs are sent to the $GetVehcileType()$ module to determine SIoV type, and for another case, it activates another module, i.e., $ThresholdCalculator()$ to calculate the threshold (X). In case 1, if any emergency vehicle, i.e., "emr" is detected on any lane, then it updates the lane priority; otherwise, it sends "No Change Signal." Similarly, for case 2, if X is less than the pre-decided threshold (th), it generates "No Change Signal," by assuming that congestion is not detected on the route; otherwise, it invokes $CongestionPreventionSignalGenerator()$ module in order to generate congestion prevention signal. The lane priority signal updates the lane priority and congestion prevention signal assists in minimizing the congestion on the route. Both signals achieve their goals by changing the phase

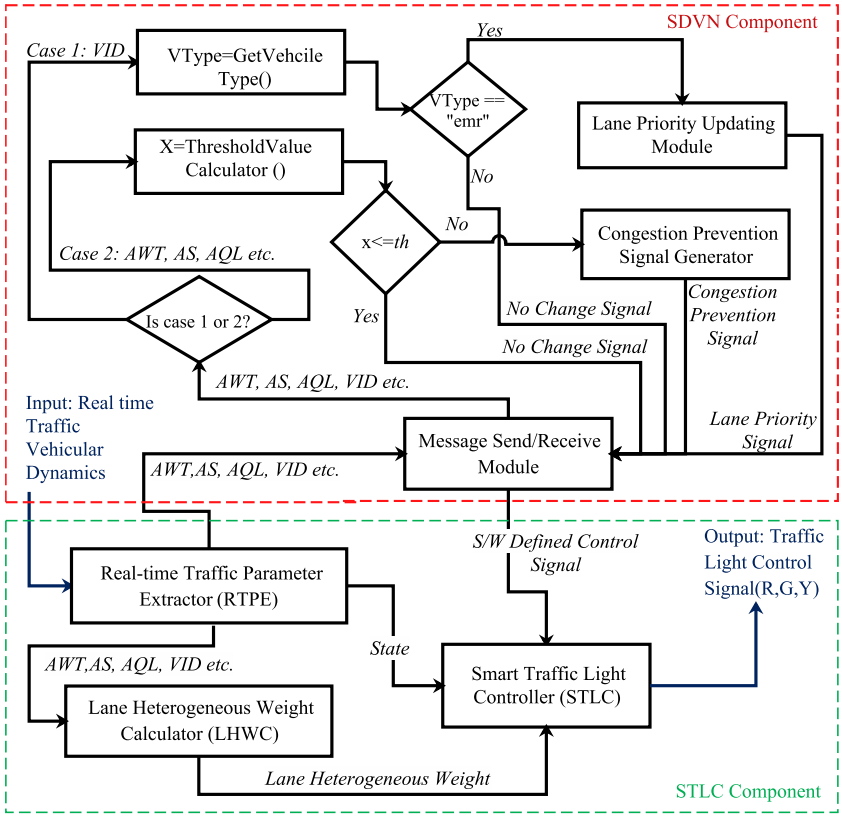


Fig. 2. Logical flow of the proposed framework.

duration of the STLC. Following this updated phase duration, STLC generates red/yellow/green signals by utilizing its multipurpose SDVN feedback-enabled novel MP algorithm.

4.3 SDNV Component

In this work, the SDVN component performs two primary functions: (1) prevent the congestion at the intersections and balance the traffic flow in the city and (2) update the lane priority based on emergency SIoV vehicle detection. For the first function, SDVN-ACS receives periodically real-time traffic parameters, i.e., AWT, AQL, and AS from all the traffic light intersections, and then congestion value (X_i) for ith junction is calculated as follows:

$$X_i = \phi NAWT_i + \chi NAQL_i - \psi \mathrm{AS}_i \tag{12}$$

where ϕ, χ, and ψ represent AWT, AQL, and AS factors, respectively. The $NAWT_i$, $NAQL_i$, and NAS_i are the normalized values for the AWT, AQL, and AS, respectively, for the ith junction which are normalized using following equation:

$$Xvar = \frac{Xvar_{curr} - Xvar_{min}}{Xvar_{max} - Xvar_{min}} \tag{13}$$

where $Xvar$ is the normalizing parameter, and min, max, and $curr$ represent the minimum, maximum, and current value for the same parameter, respectively. If $X_i < th$, SDVN controller releases a "No Change Signal" to the STLC installed at the intersection in the city, otherwise, congestion

prevention signal is estimated and sent to the STLC to control traffic flow accordingly, where, congestion prevention signal generates new phase duration to the respective STLC. The message send/receive module is responsible for all data communication between the control plane and the data plane. The sensing entities at the data plane sense the vehicular dynamics, then RSUs send it to the SDVN controller. The SDVN controller analyzes the data, generates the congestion control signal accordingly, and sends it to the respective STLC following the reverse path. As per the study [11], researchers suggested that the waiting time, AS, and queue length are the dominant parameters to quantify the congestion. Thus, three parameters are accounted for to calculate the X. For the second function, SDVN receives vehicle IDs (VID) of each SIoV and determines its type using a function, i.e., $GetVehcileType()$, if it $Vtype == emr$ then it activates *Lane Priority Updating Module* to generate a signal for upgrading the respective lane priority by a factor γ. Otherwise, the SDVN controller releases a "No Change Signal." The proposed STLC offers a relatively higher green phase duration to the respective lane by updating the lane priority so that emergency SIoV can pass the junction as soon as possible.

4.4 Algorithm

This study is dedicated to create an SIoV network for emergency vehicles and provide smother (minimal congestion and waiting time) intersection crossing service. In this context, Algorithm 1 (SDVN-MP) enhances the MP [23] algorithm by accounting vehicles heterogeneity, waiting time, and queue length to estimate the lane pressure. The MP has been proven to work in real-time on low compute-intensive micro-controller. To account for the vehicle's heterogeneity, Algorithm 2 is used. SDVN-MP periodically takes feedback from Algorithm 3 (Multi Purpose Software-Defined Control (MPSDC)) in terms of Congestion Prevention Signal. To improve the congestion, MPSDC generates a new phase-cycle duration to STLC, deployed at an intersection, and all the neighboring STLCs.

Algorithm 1. It begins by taking certain inputs, as shown in Algorithm 1. Initially, Algorithm 1 is deployed at each (jth) traffic light junction. The $Step_j$ and total cyclic phase duration (T_j) are initialized to 0 and 60 seconds, respectively, where $1 \leq j \leq N$. Before going to further simulation steps, after T_j number of timestamps, $Wmatrix_j$ is calculated for jth junction, to obtain this, $WeightCalculation(N, VID_j)$ is called. The $Wmatrix_j$ is the set of weights for all the vehicles presented on all the lanes at jth junction.

Based on $Wmatrix_j$, AQL ($\text{AQL}j, k(t_j)$) and AWT ($\text{AWT}_{j,k}(t_j)$) are estimated for kth lane of jth junction timestamp, i.e., t_j. The AQL and AWT are set with equal weightage using $\alpha = 0.5$ and $\beta = 0.5$, respectively. Using this, an effective pressure ($X_{j,k}$) is measured for each lane at the respective intersection. However, the values of α and β can be altered as per application requirements, where, $\alpha + \beta == 1$. Based on $X_{j,k}$, phase duration for each lane is estimated, and based on the lane pressure, phase duration is distributed for each lane. The maximum pressure lane initiates with the green phase duration, and the remaining lanes at the respective junction are set to red phase for the same duration at the current timestamp. Before switching between red to green or vice versa, each lane traffic light uses 3 seconds for the yellow signal [18]. For this scenario, initial cyclic phase duration is set to 60 seconds [18], however, these values can be changed as per application demand. Therefore, after every 60 seconds, traffic parameters are sent to CS using $SendMsgToCS(\text{AWT}_j, \text{AQL}_j, \text{AS}_j, T_j(t_{pre}))$, based on these parameters, SD Control module calculates new T_i that is received by TLC, i.e., $ReceiveMsgFromCS(T_j(t_{new}))$ at jth junction. Same procedure is repeated for the number of simulation steps.

Algorithm 2. The objective of the function, i.e., $WeightCalculation(N, VID)$ is to create a $Wmatrix$ for the respective junction to account its vehicles heterogeneity. Based on its type of vehicles (using $vehicle.getType(VID)$ function to get ID), constants, i.e., $Police = W_p$, $Bus = W_b$, $truck = W_t$,

Algorithm 1 SDVN-MP

Input: Total Number of junctions (J), Total Number of Vehicles (N), Vehicular Network of the city.
Output: Traffic light control signal (Green/Yellow/Red).
Notation:
T_j: Phase duration of jth junction
r_j: Number of roads intersecting at jth junction
$Wmatrix_j$: Weight Matrix for jth junction
Initialize:
$\alpha \leftarrow 0.5$; ▷ Queue length factor
$\beta \leftarrow 0.5$; ▷ Waiting time factor

1: **for** $j \in J$ **do**
2: Initialize: $T_j \leftarrow 60$, $Step_j \leftarrow 0$, $j \leftarrow 0$, $k \leftarrow 0$, $t_j \leftarrow 0$ ▷ timestamp for $j-th$ junction;
3: **while** $Step_j < SimulationSteps$ **do**
4: **if** $(t_j \ \% \ T_j) == 0$ **then**
5: $VID_j = getVehicleList(r_j)$
6: $Wmatrix_j = WeightCalculation(N, VID_j)$;
7: **for** $k \in r_j$ **do**
8: $AQL_{j,k}(t_j) = \sum_{i \in N_{j,k}} \left(\frac{Wmatrix_{j,k}(V_i)}{N_{j,k}} \right)$;
9: $AWT_{j,k}(t_j) = \sum_{i \in N_{j,k}} \frac{getWaitingTime(V_i)}{N_{j,k}}$;
10: $X_{j,k} = \alpha AQL_{j,k}(t_j) + \beta AWT_{j,k}(t_j)$;
11: **end for**
12: **for** $k \in r_j$ **do**
13: $SetGreenPhase(laneID_k, \frac{X_{j,k} \times T_i}{\sum_{i \in N_{j,k}} X_{j,k}})$;
14: $SetRedPhase(laneID_l, \frac{X_{j,k} \times T_i}{\sum_{i \in N_{j,k}} X_{j,k}})$;
▷ where, $1 \leq l \leq r_j$, and $l \neq k$
15: $SetYellowPhase(laneIDn_j, 3)$;
16: **end for**
17: $SendMsgToCS(AWT_j, AQL_j, AS_j, T_j(t_{pre}))$;
18: MPSDC (J,AWT_j, AQL_j, AS_j, $T_j(t_{pre})$);
19: T_j= $ReceiveMsgFromCS(T_j(t_{new}))$;
20: **end if**
21: t_j+ = 1;
22: $Step_j$+ = 1
23: **end while**
24: **end for**

$Emergency = W_e$, $car = W_c$, $Motorcycle = W_m$, $thers = W_o$) are appended to the $Wmatrix$ for the respective vehicle. In this work, "*Ambulance*" and "$fire_brigade$" are considered as the emergency vehicles. The constants values are set based on experimental analysis; however, these values can be set as per the requirements of the application. $Fire_brigade = 5$ are fixed. However, these weights can be changed as per the application's priority requirements.

Algorithm 3. MPSDC activates once it receives several traffic parameters *i.e.*, AWT_j, AQL_j, AS_j, $T_j(t_{pre})$ from jth STLC. In line 1, based on our recent study [17], th is set to 0.6 value. Following the same framework [17], waiting time is given relatively higher weight-age, and queue length and speed are given equal weight-age during congestion estimation, as these values are more

Algorithm 2 Weight Calculation

Input: Number of vehicle(N), Vehicle IDs (VID)
Output: $Wmatrix$ of the respective junction
Constants: $W_p, W_b, W_t, W_e, W_o, W_c, W_m$ ▷ Heterogeneous Weight factor

1: **for** $i \in N$ **do**
2: $l = vehicle.getTypeID(VID_i)$
3: **if** $l == Police$ **then**
4: $Wmatrix.append(W_p)$
5: **else if** $l == Bus$ **then**
6: $Wmatrix.append(W_b)$
7: **else if** $l == Car$ **then**
8: $Wmatrix.append(W_c)$
9: **else if** $l == Truck$ **then**
10: $Wmatrix.append(W_t)$
11: **else if** $l == Ambulance$ or $Fire_brigade$ **then**
12: $Wmatrix.append(W_e)$
13: **else if** $l == Motorcycle$ **then**
14: $Wmatrix.append(W_m)$
15: **else**
16: $Wmatrix.append(W_o)$
17: **end if**
18: **end for**
19: return $Wmatrix$

promising; therefore, in lines 2–4, ϕ, χ, and ψ are set to 0.4, 0.3, and 0.3 values, respectively. However, corresponding values can be changed as per the application requirements. Further, γ and Γ represent the lane priority and congestion balancing factors that are to be decided based on experimental analysis. The γ indicates by which factor lane priority should be increased to achieve the required/acceptable performance.

In lines 6–8, CS receives the traffic parameters from each jth TLC. In lines 10–14, the respective lane priority is increased by multiplying with the lane priority factor γ, if any lane is observed with the emergency vehicle (V^{emr}). The value of γ is decided based on experimental test cases. Next, for the jth TLC, $NAWT_j$, $NAQL_j$, and NAS_j are calculated, based on these, the expected congestion value (C_j) is estimated as shown in line 18. If $C_j \geq th$ then $T_j(t_{new})$ is increased by Γ factor, where $T_j(t_{new})$ represents new cycle phase duration for jth TLC, and $T_j(t_{new})$ for its ith neighboring TLCs is decreased by Γ factor, where, $1 \leq i \leq |Neighbors|$. Finally, the respective $T_j(t_{new})$ is sent to jth TLC where it is to be used for the subsequent iteration.

5 Simulation Study

The performance behavior of SDVN-MP is rigorously analyzed through an extensive simulation study. The simulation is prepared on the Indian city OpenStreetMap utilizing SUMO simulator version 1.9.1 [15]. To analyze the effectiveness, several important performance metrics, i.e., AWT, AS, AQL, and ATP are taken into account. Further, the proposed SIoV framework is also implemented and compared with the several TLCs, i.e., **Fuzzy Controller Algorithm for Traffic Light (FCTL)** [13], **Deep Q Learning Network Algorithm (DQN)** [21], **Max Pressure Algorithm (MPA)** [23], and Uniform [17] to analyze its comparative performance.

Algorithm 3 MultiPurposeSoftwareDefinedControl(MPSDC)

Input: J, AWT_i, AQL_i, AS_i, $T_j(t_{pre})$
Output: Updated phase duration for each Junction T_i.

$th \leftarrow 0.6$; ▷ Expected congestion threshold value
$\phi \leftarrow 0.4$; ▷ Waiting time factor
$\chi \leftarrow 0.3$; ▷ Queue length factor
$\psi \leftarrow 0.3$; ▷ Speed factor
γ, Γ ▷ lane priority, congestion balancing factors
for $j \in J$ **do**
  ReceiveMsgFromTLC(AWT_j, AQL_j, AS_j, $ATh_j(t_{pre})$);
end for
for each $j \in J$ **do in parallel**
  for $k \in r_j$ **do**
    if $V^{emr} \in vehicleType(VID_{j,k})$ **then**
      $AQL_{j,k}(t_j) = \gamma \times AQL_{j,k}(t_j)$;
    end if
  end for
  $NAWT_j = \frac{AWT_j - AWT_{min}}{AWT_{max} - AWT_{min}}$;
  $NAQL_i = \frac{AQL_j - AQL_{min}}{AQL_{max} - AQL_{min}}$;
  $NAS_j = \frac{AS_j - AS_{min}}{AWT_{max} - AS_{min}}$;
  $C_j = \phi NAWT_j + \chi NAQL_j - \psi NAS_i$;
  if $th \leq C_j$ **then**
    $T_j(t_{new}) = T_j(t_{pre}) + \Gamma \times T_j(t_{pre})$;
    for $i \in Neighbour_j$ **do**
      $T_i(t_{new}) = T_i(t_{pre}) - \Gamma \times T_i(t_{pre})$;
    end for
  end if
  SendMsgToTLC($T_j(t_{new})$);
end for

5.1 Simulation Parameters and Methodology

All the proposed SIoV framework modules are coded in Python programming, and the SIoV's vehicular network simulation is prepared in SUMO. An illustration of the working map generated with SUMO is given in Figure 3. The OpenStreetMap of Gwalior, an Indian city, is downloaded as the.osm file [15]. For the simulation, the same.osm file is converted into a vehicular network of road lanes, traffic light junctions/intersections, and vehicular traffic by utilizing the SUMO conversion tool, i.e., netconvert.

For the SIoV, a unique ID is assigned to each vehicle. In order to detect these vehicles, each lane is equipped with detectors. In particular, this work considers emergency vehicles (ambulance, fire-brigade, so on) as the SIoV. However, this framework is open to changing the type of SIoV as per the application requirements.

Traffic Control Interface (TRACI) is utilized to provide the programming control interface between Python and SUMO simulation for controlling the SUMO traffic behavior via python implementation, which is one of the modules of the SUMO. Using the TRACI module, SUMO and Python integration effectively access and control the vehicular traffic flow through Python-written algorithmic code. Table 2 represents the default SUMO simulation parameters. The randTrips.py script

Table 2. SUMO Simulation Traffic Parameters

SUMO Simulation Parameters	Value
Length of car	4.5 m
Length of bus/truck	7 m
Max. lane speed	20 m/s
Max. speed of car	15 m/s
Max. speed of bus/truck	10 m/s
Other vehicle's max speed	13.9 m/sec^2
Max. acceleration of car	2.5 m/sec^2
Max. acceleration of bus/truck	0.8 m/sec^2
Max. acceleration of other vehicles	1.8 m/sec^2
Max. acceleration of other vehicles	1.8 m/sec^2
Max. de-acceleration of all vehicles	4.0 m/sec^2
Driving proficiency(σ)	0.5
Detector frequency	60
Min. gap between two vehicles	2 m
Maximum number of lanes	4
Minimum number of lanes	2
Road lane type	Unidirectional
BD parameter n	3
Number of traffic light junctions	16
Latitude of the map	26°9′42.0264″N
Longitude of the map	78°4′21.3024″E

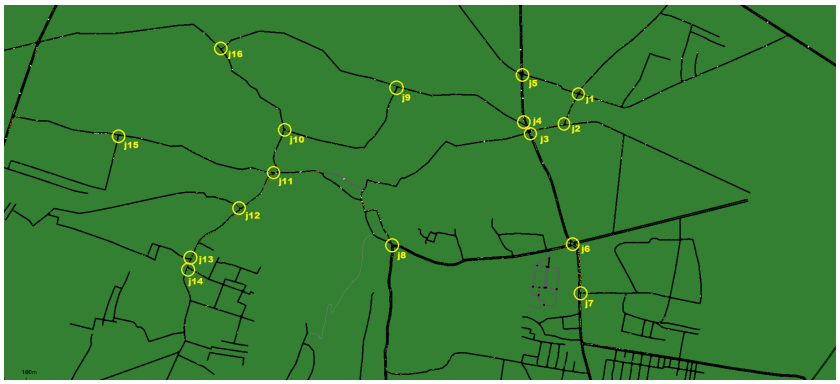


Fig. 3. An illustration of the map used for the simulation study.

is used to generate the traffic on the vehicular road lane network drawn from the OpenStreetMap in SUMO simulation. In general, this script has two ways to generate the traffic: (1) an option –period <FLOAT> (default 1) to control the arrival rate, by default, this script generates vehicles with a constant period and an arrival rate of 1/period per second, where less than 1 value represents the multiple vehicles' arrival in 1 second and (2) the exact arrival rate can also be randomized with the **Binomial Distribution (BD)** having parameter n which denotes the maximum number of simultaneous arrivals. Considering the real traffic scenario during the rush-hours, this work utilizes second way, i.e., BD to generate traffic on the road lanes. To do this, the value of n is set to 3, and the

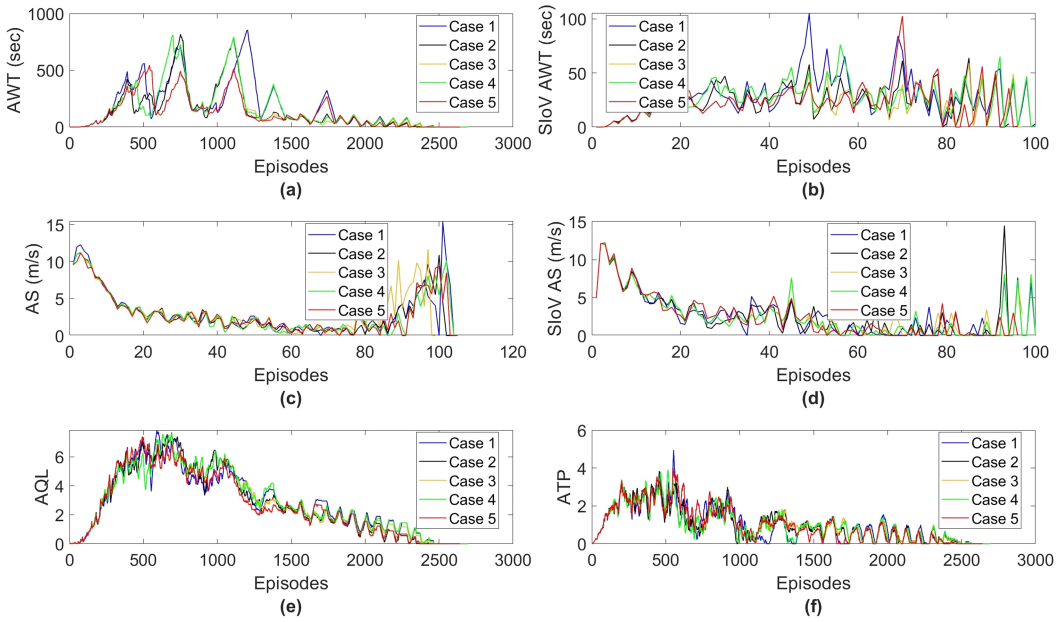


Fig. 4. Priority weights parameter tuning comparison over (a) AWT, (b) SIoV AWT, (c) AS, (d) SIoV AS, (e) AQL, and (f) ATH.

same is incorporated in the SUMO for experimentation. Hence, the vehicle's generation probability is set in the randTrips.py script accordingly.

The duration for the yellow signal is kept constant at 3 seconds which is considered the transition signal red to green or vice versa.

For the congestion management, the minimum and maximum values of the proposed function, as shown in Equation (12), are in the range of (−1, 2). If the sum of $NAWT$ and $NAQL$ is 0, and NAS is at its maximum value (1), it achieves the lowest (−1) as it shows congestion-free traffic flow. It logically implies that if the value of the proposed function (Equation (12)) is less than 0, then there is no congestion during the traffic flow. However, if the value exceeds 0, then the possibility of congestion arises, and it can be a maximum 2 considering the case $NAWT + NAQL = 2$ and $NAS = 0$ similar to [17], this signifies the traffic jam as shown in Figure 6. Based on our recent study [17], it is observed that the traffic flow is relatively smoother throughout the simulation episodes if the suggested th congestion value is set to 0.6. If it goes higher than 0.5, then results are getting worse. Thus, the suggested threshold value is 0.6. However, according to the application's requirement, this threshold value can be set to a higher or lower side.

5.2 Results and Discussions

This sub-segment portrays a detailed investigation of SDVN-MP's trial discoveries and equivalent examinations with other related works.

5.2.1 Performance Behavior of SDVN-MP over Various Parameter Settings. The performance of SDVN-MP is dissected for different execution measurements by directing a bunch of investigations on different parameter settings. A three-stage analysis is shown in Figures 4–6 and Table 3 over six performance measures. In the first stage, the priority weights of the vehicles were varied, as shown

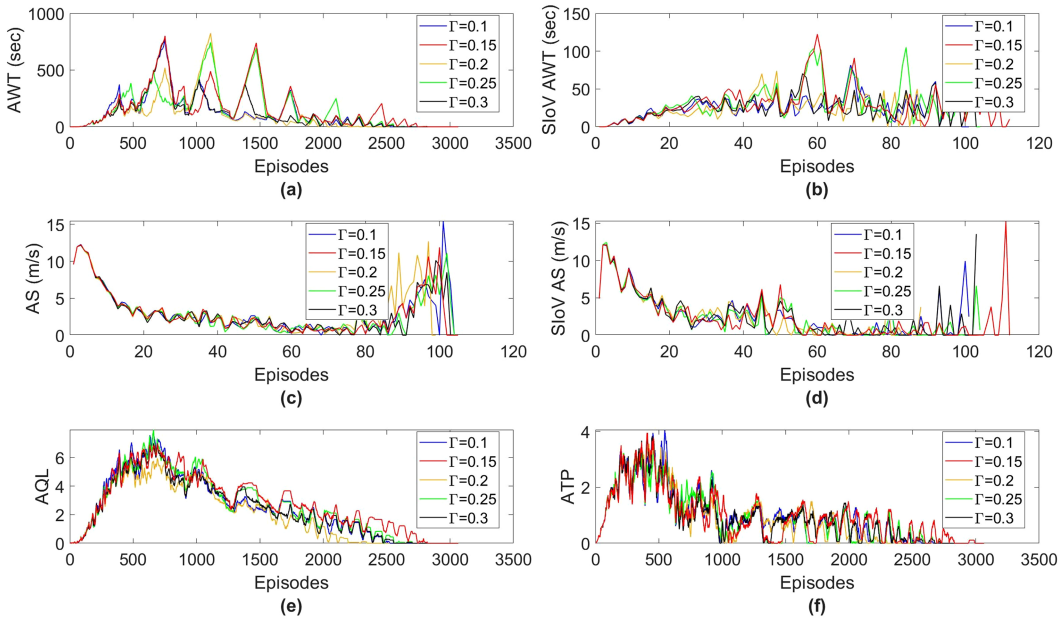


Fig. 5. Congestion balancing factor parameter tuning comparison over (a) AWT, (b) SIoV AWT, (c) AS, (d) SIoV AS, (e) AQL, and (f) ATH.

in Table 3, and the priority weights parameter tuning quantitative comparison is shown in Table 4. The following observations are inferred from Table 4 and Figure 4:

The red curve corresponding to case 5 of Table 3 in Figure 4(a) has the best AWT values in comparison to other priority weight scenarios. A lower value of AWT is preferred. Case 5 has an improvement of 30.4%, 12.7%, 18.5%, and 23.1% over case 1, case 2, case 3, and case 4, respectively. In terms of SIoV AWT, where lower values are preferred, it can be seen in Figure 4(b), case 5 outperforms others with an improvement of 23.4%, 0.5%, 6.5%, and 15.2% over case 1, case 2, case 3, and case 4, respectively. Case 5 has the edge over cases 1–4, with an improvement of 1.6%, 3.8%, 1.3%, and 1%, respectively, in terms of AS. Higher values of AS are preferred. It can be seen in Figure 4(c) that case 5 curve is higher than the other curves. In terms of SIoV AS too, where higher values are preferred, it can be seen in Figure 4(d) that case 5 outperformed the cases 1–4 with an improvement of 7.8%, 4.8%, 2.8%, and 5.3%, respectively. Lower values of AQL are preferred. From Figure 4(e), it can be deduced that case 5 results are superior. The improvement over case 1, case 2, case 3, and case 4 is 6.74%, 4%, 6%, and 8.1%, respectively. The higher the ATP better is the result. As seen from Figure 4(f), the red curve corresponding to case 5 has better performance than cases 1–4 with an improvement of 0.1%, 1.8%, 0.15%, and 0.65%, respectively. The results are significantly improved by considering case 5 since higher weight to the priority/SIoV vehicles encourages the STLC to give a larger green phase duration to pass vehicles smoothly. Further, due to a high number of priority/SIoV vehicles, the STLC decisions significantly affect the SIoV vehicles and all vehicles' performance.

Carrying forward the case 5 priority weights of Table 3, in stage 2, the congestion balancing factor, Γ, is varied over different values (0.1, 0.15, 0.2, 0.25, and 0.3), as shown in Table 4 and Figure 5. From the results in Table 4 and Figure 5, following observations are inferred: The red curve corresponding to $\Gamma = 0.15$ in Figure 5(a) has the best AWT values in comparison to other Γ values.

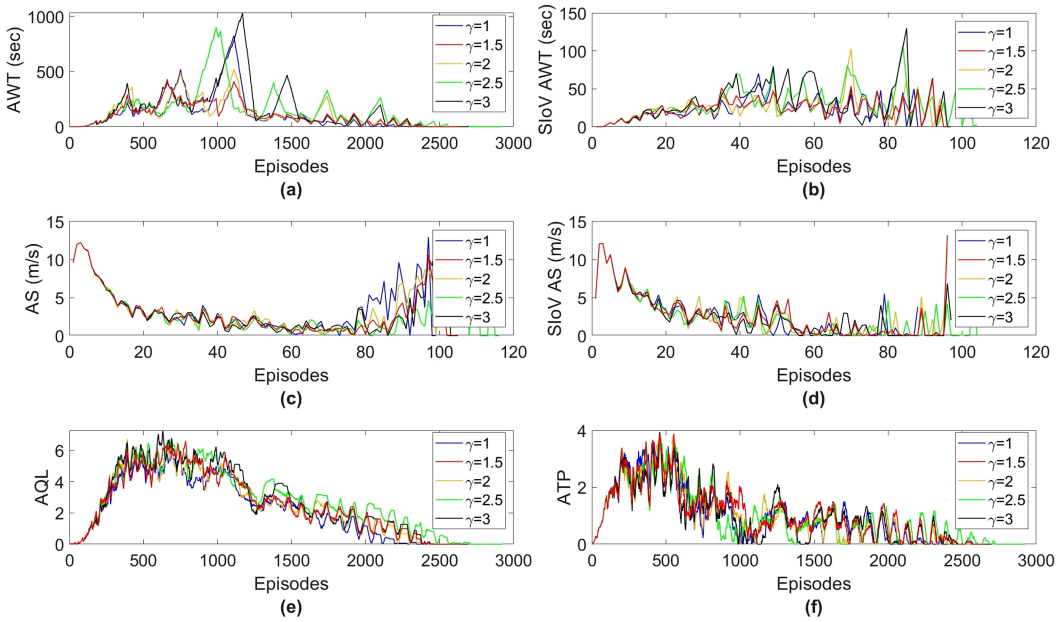


Fig. 6. Lane priority parameter tuning comparison over (a) AWT, (b) SIoV AWT, (c) AS, (d) SIoV AS, (e) AQL, and (f) ATP.

Table 3. Priority Weights Setting of Vehicles

Vehicles	Case 1	Case 2	Case 3	Case 4	Case 5
Police	10	1	10	5	5
Bus	5	1	5	5	3
Truck	5	1	5	5	3
Motorcycle	1	1	1	1	1
Ambulance	10	1	10	10	15
Fire	10	1	10	10	10
Others	3	1	1	1	2

Lower value of AWT is preferred. Red curve for $\Gamma = 0.15$ has an improvement of 14.6%, 38.9%, 34.2%, and 13.1% over curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$, and $\Gamma = 0.3$, respectively. In terms of SIoV AWT, where lower values are preferred, it can be seen in Figure 5(b), the red curve corresponding to $\Gamma = 0.15$ outperforms others with an improvement of 19.1%, 39.9%, 37.2%, and 20.1% over curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$, and $\Gamma = 0.3$, respectively. The red curve corresponding to $\Gamma = 0.15$ has an edge over curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$, and $\Gamma = 0.3$, with an improvement of 10%, 18.4%, 13.8%, and 15.7%, respectively, in terms of AS. Higher values of AS are preferred. It can be seen in Figure 5(c) that red curve of $\Gamma = 0.15$ is higher than the other curves. In terms of SIoV AS too, where higher values are preferred, it can be seen in Figure 5(d) that the red curve corresponding to $\Gamma = 0.15$ outperformed the curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$ and $\Gamma = 0.3$ with an improvement of 6.7, 9.7, 2.6, and 10% respectively. Lower values of AQL are preferred. From Figure 5(e), it can be deduced that the red curve corresponding to $\Gamma = 0.15$, results are better. The improvement over curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$ and $\Gamma = 0.3$ is 10%, 29%, 21.9%, and 15.3%, respectively. Higher the ATP better is the result. As seen from Figure 5(f), the red curve corresponding to $\Gamma = 0.15$ has

Table 4. Parameter Tuned Quantitative Evaluation of All Three Stages of the Proposed Model Over SAWT, SSAWT, SAS, SSAS, SAQL, and SATP

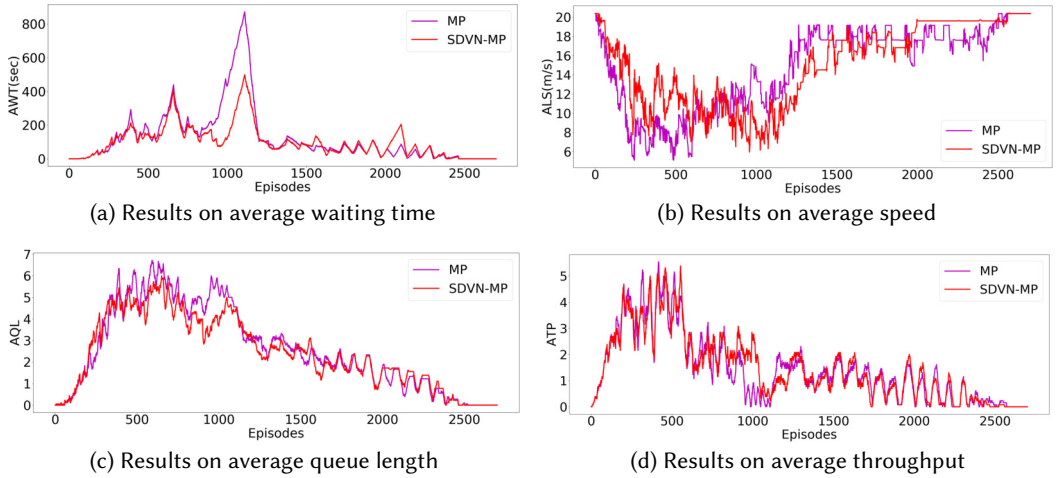
Stage	Scenario	SAWT	SSAWT	SAS	SSAS	SAQL	SATP
Stage 1	Case 1	4.81	2.89	3.79	2.40	7.65	2.71
	Case 2	3.83	2.22	3.71	2.47	7.43	2.67
	Case 3	4.10	2.37	3.80	2.52	7.60	2.71
	Case 4	4.35	2.61	3.81	2.46	7.76	2.70
	Case 5	3.35	2.21	3.85	2.59	7.13	2.71
Stage 2	$\Gamma = 0.1$	3.58	2.51	4.42	2.49	7.47	2.73
	$\Gamma = 0.15$	3.06	2.02	4.86	2.66	6.12	2.91
	$\Gamma = 0.2$	5.01	3.37	4.10	2.42	8.63	2.63
	$\Gamma = 0.25$	4.65	3.22	4.27	2.59	7.84	2.84
	$\Gamma = 0.3$	3.52	2.53	4.20	2.42	7.23	2.78
Stage 3	$\gamma = 1$x	3.06	2.22	4.06	2.44	7.18	2.60
	$\gamma = 1.5$x	2.81	2.03	4.34	2.56	6.12	2.84
	$\gamma = 2$x	3.11	2.17	4.21	2.51	6.30	2.62
	$\gamma = 2.5$x	4.27	2.93	3.99	2.53	7.00	2.74
	$\gamma = 3$x	4.50	2.86	4.30	2.50	6.57	2.61

SAWT, Sum of episode's average waiting time (in times E+5 sec); SSAWT, sum of episode's SIoV average waiting time (in times E+3 sec), SAS, sum of episode's average speed (in times E+4 m/sec); SSAS, sum of episode's SIoV average speed (in times E+2 m/sec); SAQL, sum of episode's average queue length (in times E+3); SATP, sum of episode's average throughput (in times E+3).

better performance than curves for $\Gamma = 0.1$, $\Gamma = 0.2$, $\Gamma = 0.25$, and $\Gamma = 0.3$ with an improvement of 6.5%, 10.5%, 2.3%, and 4.5%, respectively.

For this stage, the results are better when $\Gamma = 0.15$ is considered. The reason for the same is that it is a balanced factor to control the congestion at the intersection. Since, with this factor, inflows of neighboring intersections are minimized and outflow of the congested intersection is maximized. If we go with a higher factor than $\Gamma = 0.15$, other vehicles waiting for time and queue length are increased at the adjacent intersections, which deteriorates the performance w.r.t. all the vehicles, including SIoV.

Further in stage 3, keeping weights as per case 5 of Table 3 and $\Gamma = 0.15$, the lane priority, γ, parameter is varied on different values (1x, 1.5x, 2x, 2.5x, 3x). The stage 3 results of SDVN-MP can be seen in Figure 6 and Table 3. From the results, the following can be inferred: the red curve corresponding to $\gamma = 1.5$x in Figure 6(a) has the best AWT values in comparison to other γ values. Lower value of AWT is preferred. Red curve for $\gamma = 0.15$ has an improvement of 8.3%, 9.8%, 34.2%, and 37.7% over curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x, and $\gamma = 3$x respectively. In terms of SIoV AWT, where lower values are preferred, it can be seen in Figure 6(b), the red curve corresponding to $\gamma = 1.5$x outperforms others with an improvement of 8.6%, 6.6%, 30.9%, and 28.9% over curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x and $\gamma = 3$x, respectively. The red curve corresponding to $\gamma = 1.5$x has an edge over curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x, and $\gamma = 3$x, with an improvement of 6.8%, 3.2%, 8.9%, and 0.86%, respectively, in terms of AS. Higher values of AS are preferred. It can be seen in Figure 6(c) that red curve of $\gamma = 0.15$ is higher than the other curves. In terms of SIoV AS too, where higher values are preferred, it can be seen in Figure 6(d) that the red curve corresponding to $\gamma = 1.5$x outperformed the curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x, and $\gamma = 3$x with an improvement of 5%, 1.9%, 1.1%, and 2.3%, respectively. Lower values of AQL are preferred. From Figure 6(e), it



(a) Results on average waiting time

(b) Results on average speed

(c) Results on average queue length

(d) Results on average throughput

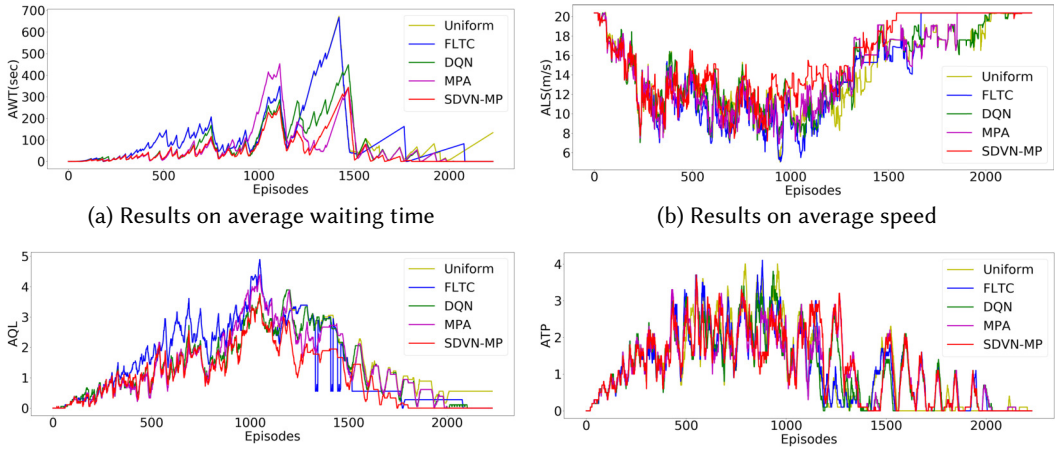
Fig. 7. Results of SDVN-MP algorithm with and without SDVN.

can be deduced that the red curve corresponding to $\gamma = 1.5$x, results are better. The improvement over curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x and $\gamma = 3$x is 14.8, 2.9, 12.6, and 6.8% respectively. Higher the ATP better is the result. As seen from Figure 6(f), the red curve corresponding to $\gamma = 1.5$x has better performance than curves for $\gamma = 1$x, $\gamma = 2$x, $\gamma = 2.5$x, and $\gamma = 3$x with an improvement of 9.3%, 8.5%, 3.9%, and 8.7% respectively.

Considering the best scenarios of stages 1 and 2, stage 3 results are further improved when $\gamma = 1.5$x is considered the lane priority factor. The red curves shown in Figure 6(a)–(f) are the results corresponding to the exemplary tuned parameters of the SDVN-MP algorithm, namely priority weights of the vehicles, congestion balancing factor, and lane priority. Based on this case study, it is observed that if one lane priority is increased, then the vehicles on the lane get higher times to pass through the intersection. However, this also affects the other remaining lanes by diminishing their green phase duration. Thus, to maintain this tradeoff, a balanced lane priority factor is required to improve the performance, i.e., $\gamma = 1.5$x observed for this case study.

5.2.2 Performance Behavior Analysis of SDVN-MP With and Without SDVN. To understand the role of SDVN, the performance comparison of the SDVN-MP algorithm with and without SDVN is presented in Figure 7. Legends in Figure 7 represent the two versions of the SDVN-MP algorithm: with SDVN denoted as SDVN-MP and without SDVN denoted as MP. Figure 7(a)–(d) depict *AWT*, *ALS*, AQL, and *ATP* performance metrics, respectively. From Figure 7(a), it is obeserved that SDVN-MP minimizes AWT by approximately 22.5% compared to MP. Similarly, Figure 7(b) shows that SDVN-MP improves *ALS* by 1.2% and reaches maximum speed first. Figure 7(c) shows SDVN-MP reducing AQL by approximately 7.2% compared to MP. Finally, Figure 7(d) shows SDVN-MP enhances *ATP* by 2.2% over MP. Based on the analysis of the results depicted in Figure 7, it is noticed that the traffic flow consistency across the simulation episodes is smoother. In brief, we can say that the proposed SDVN-MP model significantly improves the performance over MP due to the inclusion of SDVN component, which globally monitors traffic and generates congestion prevention signals to minimize the congestion.

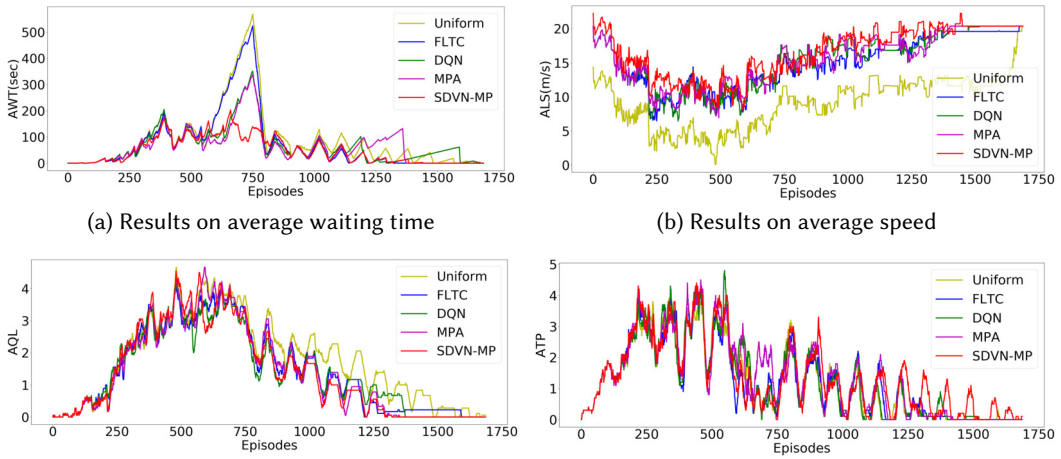
5.2.3 Performance Behavior Analysis of SDVN-MP over Other State-of-the-Art Methods. To analyze the comparative performance behavior of SDVN-MP, a set of comparative experiments with several state-of-the-art algorithms has been done. For the comparative study, DQN [21], FCTL [13], MPA



(a) Results on average waiting time (b) Results on average speed

(c) Results on average queue length (d) Results on average throughput

Fig. 8. Results on simultaneous arrivals of the vehicles (BD = 2).

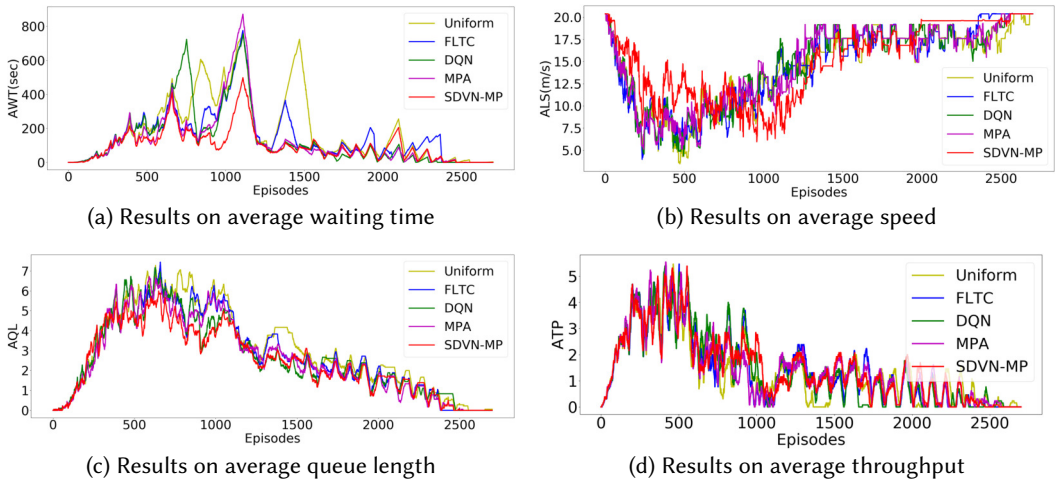


(a) Results on average waiting time (b) Results on average speed

(c) Results on average queue length (d) Results on average throughput

Fig. 9. Results on maximum number of vehicles (#vehicles = 600).

[23], and Uniform algorithms [17] are used. Figure 8 presents the results by setting BD = 2 and 1,000 vehicles in the simulation. For the AWT metric, it is observed from Figure 8(a) that performance improvement of SDVN-MP is 55.2%, 54.9%, 33%, and 26% over Uniform, FLTC, DQN, and MPA, respectively. Figure 8(b) presents the results for the **Average Lane speed (ALS)**, from the results, it is inferred that SDVN-MP is able to clear the traffic first and achieves the maximum ALS by improving its performance over Uniform, FLTC, DQN, and MPA with 33.2%, 32.9%, 30.6%, and 30.2%, respectively. In the same direction, it can be analyzed via Figure 8(c) that the SDVN-MP also improves AQL performance metric over Uniform, FLTC, DQN, and MPA by 38.4%, 29.8%, 22.9%, and 22.5%, respectively. At last, it is also inferred, from Figure 8(d), SDVN-MP outperforms Uniform, FLTC, DQN, and MPA by 8.6%, 6%, 5.5%, and 1.8% for *ATP* metric. The SDVN-MP utilizes the SDVN controller to monitor the congestion on the routes and generates practical phase duration as a



(a) Results on average waiting time

(b) Results on average speed

(c) Results on average queue length

(d) Results on average throughput

Fig. 10. Results on maximum number of vehicles (#vehicles = 1000).

congestion prevention signal to mitigate the intersection congestion. Following this, the TLC, at the intersection, optimizes the performance metrics.

To analyze the SDVN-MP algorithm's performance behavior considering the density of the vehicles, two scenarios with 600 and 1,000 vehicles are generated and results are presented in Figures 9 and 10, respectively. For AWT, results are presented in Figure 9(a), from the results, it is observed that SDVN-MP improves 44.5%, 33.1%, 24.7%, and 22.6% AWT over Uniform, FLTC, DQN, and MPA, respectively. Further, Figure 9(b) illustrates that SDVN-MP achieves 82.7%, 10.7%, 7.5%, and 6.5% improvement over Uniform, FLTC, DQN, and MPA, respectively, for *ALS* metric. Figure 9(c) presents the AQL performance improvements of SDVN-MP over Uniform, FLTC, DQN, and MPA, and it achieved 20.9%, 3.7%, 2.1%, and 1.6% improvement, respectively. Following it, Figure 9(d) shows that SDVN-MP performs 12.4%, 11.4%, 10.9%, and 1.9% better than Uniform, FLTC, DQN, and MPA, respectively, for *ATP* performance metrics. Similarly, Figure 10(a) represents 47.7%, 33.8%, 27.9%, and 22.5% SDVN-MP performance's improvement in AWT metric over Uniform, FLTC, DQN, and MPA, respectively. For the same set of experiments, Figure 10(b) shows that SDVN-MP achieves 3.8%, 2.6%, 1.4%, and 1.2% better performance over Uniform, FLTC, DQN, and MPA, respectively, for *ALS* metric. For AQL metric, results shown in Figure 10(c) present that SDVN-MP performs 19.8%, 13.5%, 8.2%, and 7.2% better than Uniform, FLTC, DQN, and MPA, respectively. The *ATP* performance improvement of SDVN-MP is shown in Figure 10(d), and the results indicate that SDVN-MP achieves 7.6%, 4.3%, 3.3%, and 2.2% better performance in comparison to Uniform, FLTC, DQN, and MPA, respectively. As an outcome of this study, the proposed SDVN-MP is able to deal with different densities of traffic and performs competitively well for AWT, AS, AQL, and *ATP* performance metrics. The reason for the same is that the SDVN-MP acts at two levels: (1) SDVN monitors the traffic globally and generates congestion prevention signal and (2) following the SDVN signal, the proposed STLC acts collaboratively to generate the control signals to smother the traffic flow and optimize the performance metrics.

6 Conclusion

This work proposes a traffic light-assisted, SDVN-enabled scheduling framework for SIoV. Due to vehicle and lane prioritization, the TLC wisely decides to pass the SIoV at the intersections. The new phase duration is estimated for the congested intersection by minimizing inflow and maximizing

the outflow from the respective intersection. To address this, an SDVN-enabled MP-based TLC is proposed, which takes the real-time traffic parameters as the inputs and generates the effective traffic light control signals. In particular, the SDVN module periodically monitors the congested traffic light intersections to update the cyclic phase duration to prevent congestion. A realistic SIoV-aware simulation test bed is developed to analyze the framework's effectiveness using an open-source simulator tool (SUMO) on an Indian city. The framework is rigorously analyzed on several performance metrics by varying testing criteria. The experimental findings demonstrate that the SDVN-MP model enhances comparative performance by 22.5–55.2%, 1.2–82.7%, 1.6–38.4%, and 1.8–12.4% for AWT, AS, AQL, and ATP metrics, respectively. The compared results with several DRL, Fuzzy, and MP-based algorithms are the evidence for the effectiveness of SDVN-MP.

The scope of this study is limited to the development of SDVN-MP for traffic light control decision making and analyzing its performance through a set of experiments in the realistic simulated environments. This study currently does not include an analysis of communication overhead and latency in real-field deployments. In future work, we plan to develop a hardware test-bed prototype to analyze communication overhead and latency across layers with an accounting for signal loss.

References

[1] Kazi Masudul Alam, Mukesh Saini, and Abdulmotaleb El Saddik. 2015. Toward social internet of vehicles: Concept, architecture, and applications. *IEEE Access* 3 (2015), 343–357.

[2] Musa Balta and Ibrahim Özcelik. 2018. Traffic signaling optimization for intelligent and green transportation in smart cities. In *Proceedings of the 2018 3rd International Conference on Computer Science and Engineering (UBMK)*. IEEE, 31–35.

[3] Prométhée Spathis, Hervé Rivano, Marcelo Dias de Amorim, Yannis Viniotis, Baron Benjamin, and Mostafa H. Ammar. 2017. Centrally controlled mass data offloading using vehicular traffic. *IEEE Transactions on Network and Service Management* 14, 2 (2017), 401–415.

[4] Pegah Nikbakht Bideh, Nicolae Paladi, and Martin Hell. 2020. Software-defined networking for emergency traffic management in smart cities. In *Proceedings of the Vehicular Ad-hoc Networks for Smart Cities: 3rd International Workshop, 2019*. Springer, Singapore, 59–70.

[5] Talal Ashraf Butt, Razi Iqbal, Sayed Chhattan Shah, and Tariq Umar. 2018. Social Internet of Vehicles: Architecture and enabling technologies. *Computers & Electrical Engineering* 69 (2018), 68–84.

[6] Yue Cao, Houbing Song, Omprakash Kaiwartya, Bingpeng Zhou, Yuan Zhuang, Yang Cao, and Xu Zhang. 2018. Mobile edge computing for big-data-enabled electric vehicle charging. *IEEE Communications Magazine* 56, 3 (2018).

[7] Zhiwen Chen, Qiong Hao, Hong Huang, and Cheng Qiao. 2023. Delay-tolerant distributed algorithms for decision-making in vehicular networks. *Asia-Pacific Journal of Operational Research* 40, 04 (2023), 2340004.

[8] Andy H. F. Chow, Rui Sha, and Shuai Li. 2019a. Centralised and decentralised signal timing optimisation approaches for network traffic control. *Transportation Research Procedia* 38 (2019), 222–241.

[9] Andy H. F. Chow, Rui Sha, and Ying Li. 2019b. Adaptive control strategies for urban network traffic via a decentralized approach with user-optimal routing. *IEEE Transactions on Intelligent Transportation Systems* 21, 4 (2019), 1697–1704.

[10] Kai-Fung Chu, Albert Y. S. Lam, and Victor O. K. Li. 2021. Traffic signal control using end-to-end off-policy deep reinforcement learning. *IEEE Transactions on Intelligent Transportation Systems* 23, 7 (2021), 7184–7195.

[11] Yi Jiang. 2001. Estimation of Traffic Delays and Vehicle Queues at Freeway Work Zones. (September 2001) Retrieved from https://journals.sagepub.com/doi/abs/10.3141/1657-02

[12] Dheeraj Jutury, Neetesh Kumar, Anuj Sachan, Yash Daultani, and Naveen Dhakad. 2023. Adaptive neuro-fuzzy enabled multi-mode traffic light control system for urban transport network. *Applied Intelligence* 53, 6 (2023), 7132–7153.

[13] Mansoure Kafash, M. Bagher Menhaj, M. Javad Motahari Sharif, and Ali Maleki. 2013. Designing fuzzy controller for traffic lights to reduce the length of queues in according to minimize extension of green light time and reduce waiting time. In *Proceedings of the 2013 13th Iranian Conference on Fuzzy Systems (IFSC)*. IEEE, 1–6.

[14] Reena Kasana, Sushil Kumar, Omprakash Kaiwartya, Rupak Kharel, Jaime Lloret, Nauman Aslam, and Tong Wang. 2018. Fuzzy-based channel selection for location oriented services in multichannel VCPS environments. *IEEE Internet of Things Journal* 5, 6 (2018), 4642–4651.

[15] Daniel Krajzewicz, Jakob Erdmann, Michael Behrisch, and Laura Bieker. 2012. Recent development and applications of SUMO-Simulation of Urban MObility. *International Journal on Advances in Systems and Measurements* 5, (2012), 3–4.

[16] Neetesh Kumar, Rashmi Chaudhry, Omprakash Kaiwartya, Neeraj Kumar, and Syed Hassan Ahmed. 2020a. Green computing in software defined social internet of vehicles. *IEEE Transactions on Intelligent Transportation Systems* 22, 6 (2020), 3644–3653.

[17] Neetesh Kumar, Sarthak Mittal, Vaibhav Garg, and Neeraj Kumar. 2021. Deep reinforcement learning-based traffic light scheduling framework for SDN-enabled smart transportation system. *IEEE Transactions on Intelligent Transportation Systems* 23, 3 (2021), 2411–2421. DOI: https://doi.org/10.1109/TITS.2021.3095161

[18] Neetesh Kumar, Syed Shameerur Rahman, and Navin Dhakad. 2020b. Fuzzy inference enabled deep reinforcement learning-based traffic light control for intelligent transportation system. *IEEE Transactions on Intelligent Transportation Systems* 22, 8 (2020), 4919–4928.

[19] Neetesh Kumar and Deo Prakash Vidyarthi. 2018. A green routing algorithm for IoT-enabled software defined wireless sensor network. *IEEE Sensors Journal* 18, 22 (2018), 9449–9460.

[20] Song Li, Bowen Wang, Shenshen Qian, Yanjing Sun, Xiao Yun, and Yu Zhou. 2022. Influence maximization for emergency information diffusion in social internet of vehicles. *IEEE Transactions on Vehicular Technology* 71, 2 (2022), 8768–8782. DOI: https://doi.org/10.1109/TVT.2022.3146260

[21] Xiaoyuan Liang, Xusheng Du, Guiling Wang, and Zhu Han. 2019. A deep q learning network for traffic lights' cycle control in vehicular networks. *IEEE Transactions on Vehicular Technology* 68, 2 (2019), 1243–1253.

[22] Junxiu Liu, Sheng Qin, Yuling Luo, Yanhu Wang, and Su Yang. 2022. Intelligent traffic light control by exploring strategies in an optimised space of deep Q-learning. *IEEE Transactions on Vehicular Technology* 71, 2 (2022), 5960–5970.

[23] Pedro Mercader, Wasim Uwayid, and Jack Haddad. 2020. Max-pressure traffic controller based on travel times: An experimental analysis. *Transportation Research Part C: Emerging Technologies* 110 (2020), 275–290.

[24] Muhammd Zeshan Naseer. 2016. Modeling control traffic in distributed software defined networks. Master's thesis. TRITA EE 2016:107. Retrieved from https://www.diva-portal.org/smash/get/diva2:1038710/FULLTEXT01.pdf

[25] Anuj Sachan and Neetesh Kumar. 2022. SDN control-enabled and time-quantum-driven max-pressure approach for intersection management in smart city. *IEEE Systems Journal* 17, 1 (2022), 1694–1702.

[26] Anuj Sachan and Neetesh Kumar. 2024. SDN-enabled quantized LQR for smart traffic light controller to optimize congestion. *ACM Transactions on Internet Technology* 24, 1, Article 7 (2024), 1–25.

[27] Ousmane Sadio, Ibrahima Ngom, and Claude Lishou. 2019. Design and prototyping of a software defined vehicular networking. *IEEE Transactions on Vehicular Technology* 69, 1 (2019), 842–850.

[28] Chen Sun, Shen Li, Dongpu Cao, Fei-Yue Wang, and Amir Khajepour. 2023. Tabular Learning-Based Traffic Event Prediction for Intelligent Social Transportation System. *IEEE Transactions on Computational Social Systems* 10, 3 (2023), 1199–1210.

[29] Chaogang Tang, Xianglin Wei, Mingyang Hao, Chunsheng Zhu, Rongcun Wang, and Wei Chen. 2018. Traffic signal phase scheduling based on device-to-device communication. *IEEE Access* 6 (2018), 47636–47645.

[30] Elise van der Pol. 2016. Deep reinforcement learning for coordination in traffic light control. Master's thesis. University of Amsterdam.

[31] Fei-Yue Wang, Yong Yuan, Juanjuan Li, Dongpu Cao, Lingxi Li, Petros A. Ioannou, and Miguel Ángel Sotelo. 2018. From intelligent vehicles to smart societies: A parallel driving approach. *IEEE Transactions on Computational Social Systems* 5, 3 (2018), 594–604.

[32] Tong Wang, Azhar Hussain, Lejun Zhang, and Chen Zhao. 2021. Collaborative edge computing for social internet of vehicles to alleviate traffic congestion. *IEEE Transactions on Computational Social Systems* 9, 1 (2021), 184–196.

[33] Xiaonan Wang and Yanli Li. 2019. Content retrieval based on vehicular cloud in internet of vehicles. *IEEE Transactions on Computational Social Systems* 6, 3 (2019), 582–591.

[34] Zhuoqun Xia, Xiaoxiao Mao, Ke Gu, and Weijia Jia. 2021. Two-dimensional behavior marker-based data forwarding incentive scheme for fog computing-based SIoVs. *IEEE Transactions on Computational Social Systems* 9, 5 (2021), 1406–1418. DOI: https://doi.org/10.1109/TCSS.2021.3129898

[35] Mobin Yazdani, Majid Sarvi, Saeed Asadi Bagloee, Neema Nassir, Jeff Price, and Hossein Parineh. 2023. Intelligent vehicle pedestrian light (IVPL): A deep reinforcement learning approach for traffic signal control. *Transportation Research Part C: Emerging Technologies* 149 (2023), 103991.

[36] Weiqi Yu, Weichen Bai, Liang Qi, and Wenjing Luan. 2022. State-of-the-art review on traffic control strategies for emergency vehicles. *IEEE Access* 10 (2022), 109729–109742.

[37] Zhihong Yin, Keping Yu, Xiongyan Tang, Lexi Xu, Zhenzhou Guo Zhao, Liang and Pulkit Nehra. 2022. A fuzzy logic based intelligent multi-attribute routing scheme for two-layered SDVNs. *IEEE Transactions on Network and Service Management* 19, 4 (2022), 4189–4200.

Received 25 November 2023; revised 28 February 2024; accepted 19 March 2024